

Michael Mills

+1 (207) 712-7052 | mmills6060@gmail.com | New York, NY | [linkedin.com/in/michael-a-mills/](https://www.linkedin.com/in/michael-a-mills/) | github.com/mmills6060

Skills

Typescript / React / Node.js / Python / Javascript / Django / Docker / MongoDB / Express.js / Postman / System Architecture / Git / MySQL / REST APIs / AWS / PostgreSQL

Professional Experience

VETRO

Software Engineer

Remote

September 2024 - July 2025

- #1 GitHub contributor to a flagship product that generated \$2.4 million in revenue within the first six months of launch, demonstrating impactful contributions to high-growth software projects and visions.
- Delivered full-stack functionality—data ingestion pipelines, AI/ML inference services, and React/Next.js front-ends written in Typescript—to solve client-specific problems and ship production-grade applications at speed.
 - Architected RESTful APIs handling 100,000+ daily requests with sub-100ms response times.
 - Implemented real-time data streaming using WebSockets for live network topology visualization.
 - Optimized database queries reducing load times through indexing and query refactoring.
- Developed and optimized network services and APIs written in Python/GoLang that interact with core TCP/IP networking layers.
- Worked closely with various engineering teams to implement CI/CD pipelines and optimize infrastructure performance for production deployments.
 - Automated deployment workflows using Github Actions.
 - Implemented containerized microservices architecture with Docker and AWS for scalable deployments.
- Managed and maintained Linux-based networking systems, ensuring high availability and performance.
- Collaborated with a scrum team in an Agile environment to develop, refine, and ship full-stack applications using modern frameworks (React.js, Vue.js, Express.js) and languages (Typescript, Python)

Banbury

Software Development Engineer

Remote

August 2023 - August 2024

- Built an AI farming robot that tilled, planted, maintained, grew, and harvested various crops such as tomatoes and potatoes using AI object detection, stereo visual odometry, motion planning.
 - Trained YOLOv8 models achieving 95% accuracy for crop identification and weed detection.
 - Implemented ROS2-based navigation stack with SLAM for autonomous field traversal.
 - Developed custom inverse kinematics algorithms for robotic arm manipulation.
- Engineered long-range wireless communication systems using specialized Wi-Fi chip technology to enable reliable robot control and data transmission over 4-6 mile distances in agricultural environments.
 - Implemented packet loss mitigation strategies ensuring command delivery in variable field conditions.
 - Optimized battery consumption through adaptive transmission power management.
 - Developed error correction and retry mechanisms maintaining communication reliability over extended distances.
- Contributed to Agile development and CI/CD practices by collaborating daily with senior software engineers, implementing CI/CD pipelines to streamline development and deployment processes.
- Leveraged relational databases for data management, effectively supporting AI-driven applications.
- Created an extensive API library and a Software Development Kit (SDK), facilitating future enhancements and customization, ensuring scalability and adaptability of the technology.

UBS

Registered Client Service Associate

New York, NY, USA

January 2021 - January 2023

- Advanced to a senior role within the Client Service Associate team, specializing in overseeing comprehensive compliance-related issues, including anti-money laundering protocols and financial advisor performance.
- Increased data-driven decision making by utilizing BI tools and statistics to analyze compliance trends, providing strategic advice and highlighting areas for improvement.
- Designed and maintained SQL-based data pipelines to automate compliance reporting and account audit workflows, reducing manual review time and improving data integrity across thousands of records.
- Implemented process automation scripts using Python, C++ and SQL to standardize data reconciliation, reducing operational errors and improving audit traceability.

Lighthouse Point Consulting

Software Engineer

Remote

March 2018 - January 2021

- Developed small to mid-scale applications and automation scripts in Python, JavaScript/TypeScript, and other languages as project needs required.
- Created and modified APIs, services, and integrations to connect disparate systems and enable data flow between platforms.

- Configured and customized Salesforce workflows, data models, and API integrations, improving CRM usability and system interoperability.
- Implemented and supported authentication flows, including OTP (one-time password) verification and SSO (single sign-on) setup for web applications.
- Assisted with cloud deployments (AWS, Docker, ECS) and CI/CD workflows to ensure stable delivery of short-term projects.
- Contributed to debugging and troubleshooting efforts, resolving production issues and technical blockers under tight deadlines.

Education

Northeastern University
Master's, Computer Science

January 2023 - June 2025

Gettysburg College
Bachelor's, Organization and Management Studies

August 2016 - March 2020

Research Publications and Side Projects

Banbury Workspaces

New York, NY, USA
August 2025 - Present

- Built an enterprise AI productivity platform with LangGraph-powered autonomous agents, featuring document/spreadsheet/presentation editing, Gmail/Calendar integration, zoom meeting bots, browser automation, 40+ API integrations using Next.js, TypeScript, and React.
 - Designed scalable microservices architecture handling concurrent multi-agent workflows.
 - Integrated OpenAI and Anthropic APIs.
 - Implemented OAuth 2.0 authentication flows for third-party service integrations.

Future-Proofing Computer Behavior: A Framework for Predictive Analytics and Seamless Device Integration

Remote

April 2024 - August 2024

- Utilized predictive analytics to enhance the understanding and management of device behaviors and performance metrics, including Wi-Fi upload/download speeds, GPU usage, CPU usage, and RAM usage.
- Developed and implemented a relay server strategy to enable seamless and secure communication between devices across different network environments.
- Built and trained a Long Short-Term Memory (LSTM) network model to forecast device performance metrics and facilitate informed decision-making for resource allocation.
- Analyzed the performance of predictive models using Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) to ensure high accuracy in predictions.